MTH203: Sequences and series

Tutorial 02

26/08/2025

This is a *practice test* for the first Mid-semester Examination.

Problem 1. (2 points each) Define the following terms:

(a) Totally ordered set

(b) Supremum

(c) Power set

(d) Countable set

Problem 2. (6 points) Let A be a subset of $\mathbb{R}$. Let $u = \sup(A)$ and $v = \inf(A)$. Let x be a real number and let

$$B = \{xa | x \in A\}.$$

Prove that

$$\sup(B) = \begin{cases} xu & \text{if } x \geqslant 0, \\ xv & \text{if } x < 0. \end{cases}$$

Problem 3. (5 points) Let n be a positive integer. Let R be the relation from $\mathbb{R}$ to $\mathbb{R}$, defined by

$$R = \{(x, y) \in \mathbb{R} \times \mathbb{R} | x = y^n\}.$$

Prove that R is a function if and only if n is odd.

Problem 4. (6 points) Let S be the subset of $\mathbb{R}$ that is bounded below. Prove that

$$\inf(S) = -\sup(\{-x | x \in S\}.)$$